Transfusion on-call: A haematologist's cheat sheet

Positive crossmatch but antibody screen is negative

Causes and what action to take

Causes	Actions
Antibody to a low frequency antigen (an antigen that occurs in <1% of the population, often not found on screening cells)	• Perform antibody panel • Refer to *Transfusing patients with red cell antibodies* cheat sheet for advice on how to safely transfuse the patient. • Can often find alternative RBC component to provide IAT cross-match compatible red cells
Sensitised donor red cells e.g. Donor has a positive DAT	• Perform DAT on donor red cells • Refer to *The direct antiglobulin test* cheat sheet for advice on how to safely transfuse the patient.
Passively transfused anti-A or anti-B in the context of massive haemorrhage or high dose IVIg	• Correlate with clinical history
ABO incompatibility due to incorrect ABO group of patient or donor unit	• Perform group check on recipient and donor sample
Passenger lymphocyte syndrome • Graft derived (donor) viable B lymphocytes are passively transferred at time of solid organ transplantation. • Results in the development of red cell antibodies of donor origin, including both ABO and non-ABO red cell antibodies. • The risk is dependent on amount of lymphoid tissue transplanted with organ: Liver > heart-lung > kidney-pancreas > kidney > pancreas • High titre red cell antibodies develop within 7-14 days of transplant and disappear within 1-6 months. • Presents as mild to severe, self-limiting intra or extravascular haemolysis. Acute renal failure, disseminated intravascular coagulopathy and multi organ failure have been reported. • The DAT is positive and most commonly IgG. • Treatment includes transfusion of recipient and donor compatible RCs and immunosuppression with or without therapeutic plasma exchange.	• Correlate with clinical history